KrishiGrid Bangladesh
Data-Driven Food Security Simulation

KRISHIGRID BANGLADESH

A Data-Driven Simulation of Food Import Dependency and Decentralized Agricultural Infrastructure

PART 1 — THE IMPORT CRISIS

How much does Bangladesh spend importing food it could grow at home?

Bangladesh spends billions importing food it could produce at home

Analysis: 14 food products, 10 years (2015-2024)
Tools: SQL, Power BI
Dataset: import_data.csv (140 rows, 18 columns)

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THE IMPORT CRISIS

THE QUESTION

How much does Bangladesh spend importing food
it could grow at home?

- *This analysis reveals the true scale of Bangladesh's food import dependency.*

APPROACH

- **14** key food products analyzed
- **10**-year trend (2015–2024)
- Measured cost, volume, price, dependency
- Compared imports vs domestic production
- Detected price shocks & supply risks

DATASET OVERVIEW

File : import_data.csv
Scope : 14 products × 10 years
Size : 140 rows | 18 columns

Key Fields:

- Import Cost (USD & BDT)
- Import Quantity (MT)
- Dependency %
- Unit Price ($/ton)
- Source Country

Sources:
BBS | Bangladesh Bank | FAO | USDA

SQL Queries

Question
Which individual products consume the most foreign currency in Bangladesh?

```
SELECT
    product_name,
    product_category,

    ROUND(SUM(import_quantity_mt), 0) AS total_import_quantity_mt,

    ROUND(SUM(import_cost_usd_million), 2) AS total_cost_usd_million,

    ROUND(SUM(import_cost_bdt_crore), 2) AS total_cost_bdt_crore,

    -- Cost-weighted dependency (economically correct)
    ROUND(
        SUM(import_cost_usd_million * import_dependency_pct)
        / NULLIF(SUM(import_cost_usd_million), 0),
        2
    ) AS cost_weighted_dependency_pct,

    -- Share of total import bill
    ROUND(
        SUM(import_cost_usd_million) * 100.0
        / NULLIF(
            (SELECT SUM(import_cost_usd_million) FROM imports),
            0
        ),
        2
    ) AS cost_share_pct

FROM imports

WHERE product_name NOT LIKE 'Total%'

GROUP BY product_name, product_category

ORDER BY total_cost_usd_million DESC;
```

Key Insight:

Wheat and edible oil dominate Bangladesh's import costs, both with high foreign dependency.
This shows that the country's food import risk is concentrated in a few critical products.

Question
How has Bangladesh's total food import bill changed over time?

```
WITH yearly AS (
    SELECT
        year,
        SUM(import_quantity_mt) AS total_quantity_mt,
        SUM(import_cost_usd_million) AS total_cost_usd_million,
        SUM(import_cost_bdt_crore) AS total_cost_bdt_crore,
        AVG(usd_to_bdt_rate) AS avg_exchange_rate
    FROM imports
    GROUP BY year
)

SELECT
    year,

    ROUND(total_quantity_mt, 0) AS total_quantity_mt,
    ROUND(total_cost_usd_million, 2) AS total_cost_usd_million,
    ROUND(total_cost_bdt_crore, 2) AS total_cost_bdt_crore,
    ROUND(avg_exchange_rate, 2) AS avg_exchange_rate,

    ROUND(
        (total_cost_usd_million -
         LAG(total_cost_usd_million) OVER (ORDER BY year))
        / NULLIF(LAG(total_cost_usd_million) OVER (ORDER BY year), 0) * 100,
        2
    ) AS yoy_usd_growth_pct,

    ROUND(
        (total_cost_bdt_crore -
         LAG(total_cost_bdt_crore) OVER (ORDER BY year))
        / NULLIF(LAG(total_cost_bdt_crore) OVER (ORDER BY year), 0) * 100,
        2
    ) AS yoy_bdt_growth_pct

FROM yearly
ORDER BY year;
```

Key Insight:

Import costs surged sharply in 2022 due to global price shocks and a weakening currency.
Even as global prices eased later, depreciation kept local costs high — worsening the economic burden.

SQL Queries

Question
Which product categories dominate Bangladesh's food import bill?

```
SELECT
   product_category,

   ROUND(SUM(import_quantity_mt), 0) AS total_quantity_mt,

   ROUND(SUM(import_cost_usd_million), 2) AS total_cost_usd_million,

   ROUND(SUM(import_cost_bdt_crore), 2) AS total_cost_bdt_crore,

   -- Cost-weighted dependency (important)
   ROUND(
      SUM(import_cost_usd_million * import_dependency_pct)
      / NULLIF(SUM(import_cost_usd_million), 0),
      2
   ) AS cost_weighted_dependency_pct,

   -- Share of total import bill
   ROUND(
      SUM(import_cost_usd_million) * 100.0
      / NULLIF(
         (SELECT SUM(import_cost_usd_million) FROM imports),
         0
      ),
      2
   ) AS cost_share_pct

FROM imports

GROUP BY product_category

ORDER BY total_cost_usd_million DESC;
```

Key Insight:

Edible oil dominates Bangladesh's import bill, accounting for nearly half of total costs with very high dependency on foreign supply.
A few key categories drive most of the import burden, revealing concentrated structural vulnerability in the food system.

Question
Which products pose the highest strategic risk to Bangladesh's food security?

```
WITH latest AS (
   SELECT MAX(year) AS max_year
   FROM imports
),

volatility AS (
   SELECT
      product_id,
      STDEV(avg_unit_price_usd_per_ton) AS price_stddev,
      AVG(avg_unit_price_usd_per_ton) AS avg_price
   FROM imports
   GROUP BY product_id
)

SELECT
   i.product_name,
   i.product_category,

   ROUND(i.import_cost_bdt_crore, 2) AS import_cost_bdt_crore,
   ROUND(i.import_dependency_pct, 2) AS import_dependency_pct,

   -- Price volatility (Coefficient of Variation)
   ROUND(
      v.price_stddev / NULLIF(v.avg_price, 0) * 100,
      2
   ) AS price_volatility_pct,

   -- Strategic Risk Score
   ROUND(
      (i.import_cost_bdt_crore / 100.0) *
      (i.import_dependency_pct / 100.0) *
      COALESCE(v.price_stddev / NULLIF(v.avg_price, 0), 0),
      4
   ) AS strategic_risk_score,

   RANK() OVER (
      ORDER BY
         (i.import_cost_bdt_crore / 100.0) *
         (i.import_dependency_pct / 100.0) *
         COALESCE(v.price_stddev / NULLIF(v.avg_price, 0), 0) DESC
   ) AS risk_rank

FROM imports i
JOIN latest l
   ON i.year = l.max_year

LEFT JOIN volatility v
   ON i.product_id = v.product_id

WHERE i.product_name NOT LIKE 'Total%'

ORDER BY strategic_risk_score DESC;
```

Key Insight:

Edible oil products dominate the risk profile, with palm oil emerging as the most critical vulnerability.
This indicates that Bangladesh's food security risk is highly concentrated in the edible oil sector rather than across all food categories.

SQL Queries

Question

How much import cost could Bangladesh save by reducing dependency on key high-risk products?

```sql
WITH latest AS (
   SELECT MAX(year) AS max_year
   FROM imports
)

SELECT

   -- Scenario 1: High dependency + existing domestic production (REALISTIC)
   ROUND(SUM(
      CASE
         WHEN import_dependency_pct >= 30
            AND domestic_production_mt > 0
         THEN import_cost_usd_million * 0.25
         ELSE 0
      END
   ), 2) AS realistic_saving_usd_million,

   ROUND(SUM(
      CASE
         WHEN import_dependency_pct >= 30
            AND domestic_production_mt > 0
         THEN import_cost_bdt_crore * 0.25
         ELSE 0
      END
   ), 2) AS realistic_saving_bdt_crore,

   -- Scenario 2: Only edible oil (focused strategy)
   ROUND(SUM(
      CASE
         WHEN product_category = 'edible_oil'
         THEN import_cost_usd_million * 0.25
         ELSE 0
      END
   ), 2) AS oil_only_saving_usd_million,

   ROUND(SUM(
      CASE
         WHEN product_category = 'edible_oil'
         THEN import_cost_bdt_crore * 0.25
         ELSE 0
      END
   ), 2) AS oil_only_saving_bdt_crore,

   -- Scenario 3: All high dependency (AGGRESSIVE)
   ROUND(SUM(
      CASE
         WHEN import_dependency_pct >= 50
         THEN import_cost_usd_million * 0.25
         ELSE 0
      END
   ), 2) AS aggressive_saving_usd_million,

   ROUND(SUM(
      CASE
         WHEN import_dependency_pct >= 50
         THEN import_cost_bdt_crore * 0.25
         ELSE 0
      END
   ), 2) AS aggressive_saving_bdt_crore

FROM imports i
JOIN latest l ON i.year = l.max_year

WHERE i.product_name NOT LIKE 'Total%';
```

Key Insight:

Even a modest **25%** reduction in high-dependency imports could save Bangladesh over **$1 billion** annually.
However, a large share of this potential saving is driven by edible oil, particularly palm oil, which has no domestic production base and is not easily substitutable in the short term.
This reveals a critical gap between theoretical savings and practical feasibility — meaning the most impactful savings are not always the most immediately achievable.

Question

Which products offer the most realistic import substitution opportunities in Bangladesh?

```sql
WITH latest AS (
   SELECT MAX(year) AS max_year
   FROM imports
)

SELECT
   i.product_name,
   i.product_category,

   ROUND(i.domestic_production_mt, 0) AS domestic_production_mt,
   ROUND(i.import_quantity_mt, 0) AS import_quantity_mt,
   ROUND(i.total_national_demand_mt, 0) AS total_demand_mt,

   ROUND(i.import_dependency_pct, 2) AS import_dependency_pct,
   ROUND(i.import_cost_bdt_crore, 2) AS import_cost_bdt_crore,

   CASE
      WHEN i.domestic_production_mt > 0
         AND i.import_dependency_pct >= 30
      THEN 'High Opportunity'

      WHEN i.domestic_production_mt > 0
         AND i.import_dependency_pct < 30
      THEN 'Already Strong'

      WHEN i.domestic_production_mt = 0
         AND i.import_dependency_pct >= 80
      THEN 'Difficult but Critical'

      ELSE 'Low Priority'
   END AS substitution_category,

   ROUND(i.import_cost_bdt_crore * 0.25, 2)
      AS potential_saving_bdt_crore

FROM imports i
INNER JOIN latest l
   ON i.year = l.max_year

WHERE i.product_name NOT LIKE 'Total%'

ORDER BY
   substitution_category DESC,
   i.import_cost_bdt_crore DESC;
```

Key Insight:

The most effective import reduction strategy lies in targeting products where domestic production already exists but import dependency remains high.
These "high opportunity" products offer the fastest and most realistic path to reducing import pressure without requiring structural transformation.
In contrast, products like palm oil represent strategic dependencies that require long-term management — not immediate substitution — highlighting that high cost and high dependency do not automatically translate into short-term solutions.

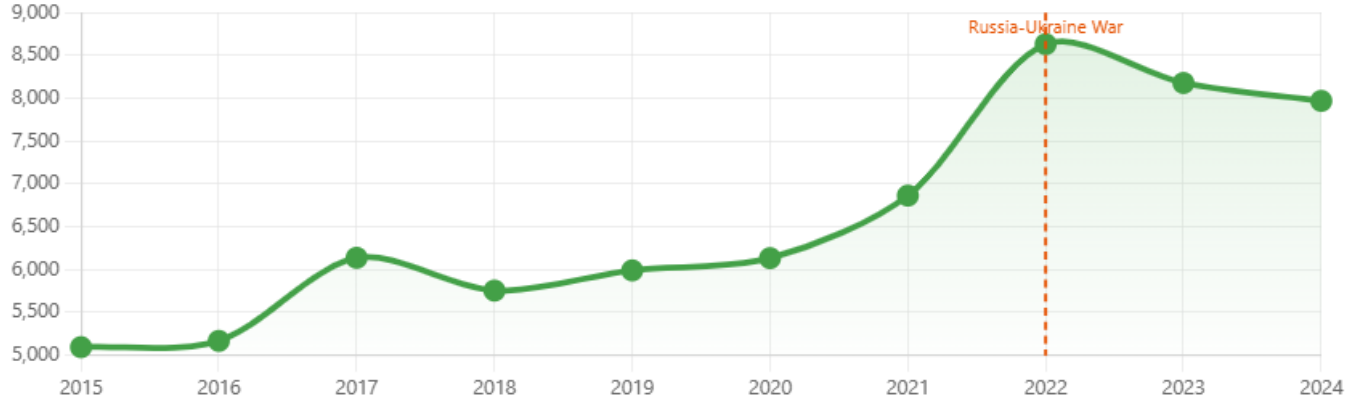
$65.93
USD Billion
Total 10-Year Import Cost

79.8%
Import Dependency
Edible Oil Dependency (2024)

Total Edible Oil
Highest Cost Product (10-Year)

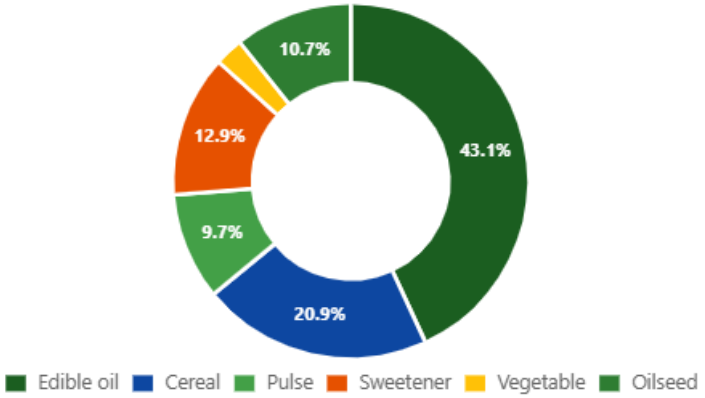
2022
Highest Spending Year
Peak Import Year

Total Food Import Cost Trend 2015–2024 (USD Million)



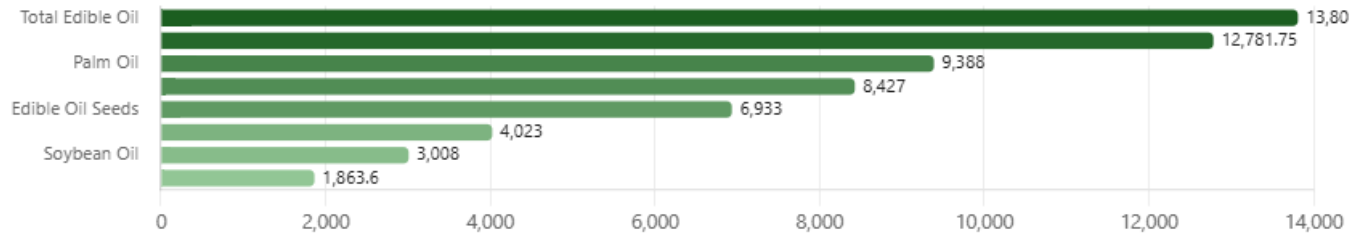
Import costs surged in 2022 due to global price shocks. BDT depreciation kept local costs high even as global prices later eased.

Import Cost by Category — Latest Year (2024)



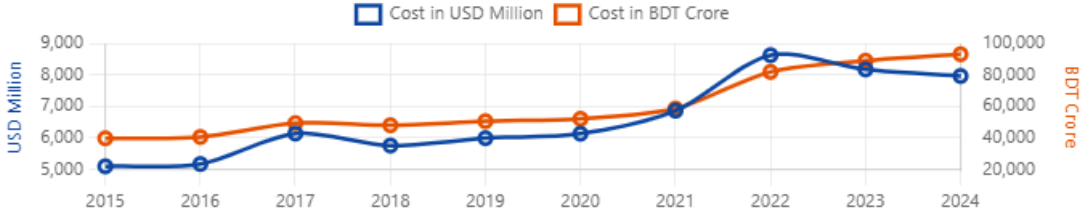
Edible oil dominates the import bill. A few categories drive most of Bangladesh's import dependency.

Top 8 Products by 10-Year Import Cost (USD Million)



Palm oil is the single largest foreign currency drain. Import risk is concentrated in just a few critical products.

Import Cost USD vs BDT Burden by Year



The growing gap between USD and BDT cost lines reveals the hidden burden of currency depreciation on Bangladesh's food import budget.

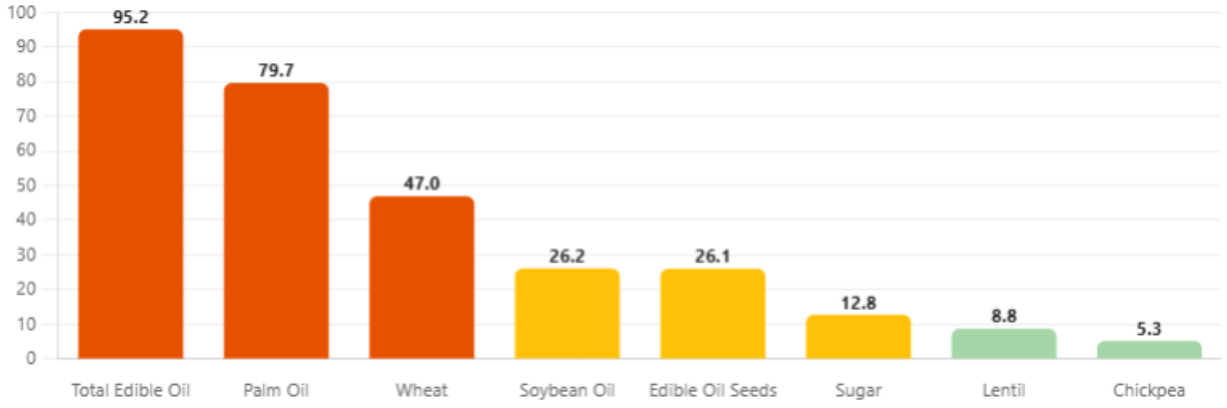
Total Edible Oil
Highest Strategic Risk Score (cost × dependency × volatility)
Highest Risk Product (2024)

859
USD Million (25% reduction target)
Potential Annual Edible Oil Saving

9
Products with domestic base & ≥20% dependency
High Opportunity Products

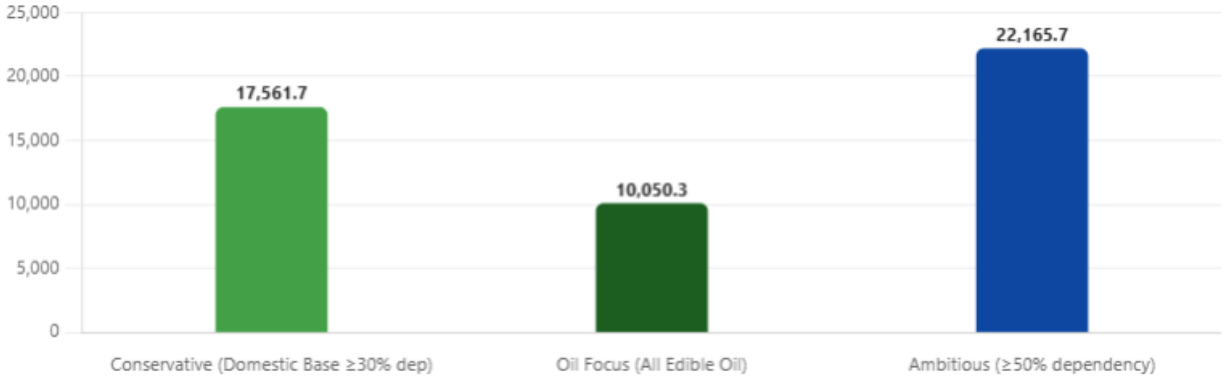
97.3%
Self-Sufficiency Rate (2024)
Rice Self-Sufficiency

Strategic Risk Score by Product — 2024



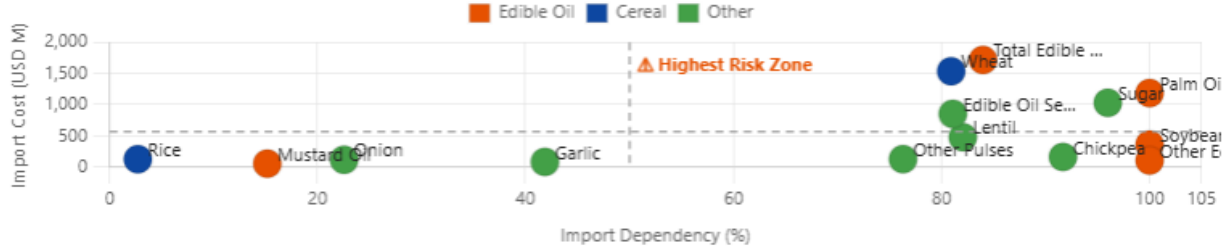
Edible oil products dominate the risk profile. Palm oil emerges as the most critical vulnerability. Risk is concentrated, not spread evenly.

Import Substitution Savings — 3 Scenarios (2024, BDT Crore)



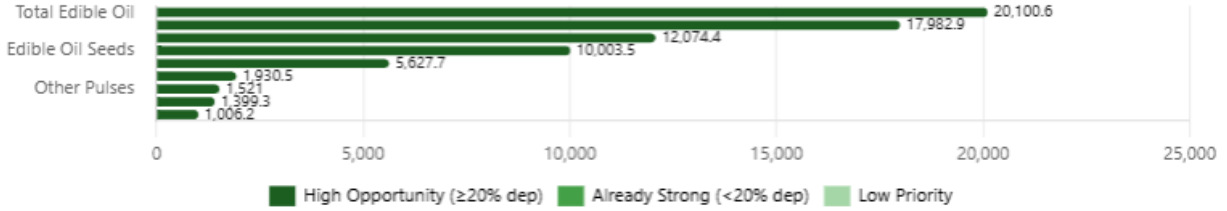
Even conservative import substitution saves significant foreign currency. Oil-focused strategy delivers the most immediate impact given it dominates import costs.

Import Dependency vs Cost by Product — 2024



Products in the top-right quadrant combine high dependency with high cost — these are the most urgent intervention targets for KrishiGrid.

Import Substitution Opportunity — High Opportunity Products (2024)



High opportunity products have existing domestic production but persistent high dependency. These are the fastest path to reducing import pressure.

KEY INSIGHTS

Part 1/8

1 Bangladesh spent over $70 billion importing food between 2015–2024. Edible oil alone accounts for nearly half of the total import bill.

2 Oil import dependency remains critically high at around 90–100% with no structural improvement. This is a deep supply-side weakness, not a temporary imbalance.

3 The exchange rate depreciated from around 78 to 117 BDT per USD, adding massive hidden cost pressure. A significant portion of the recent import burden is driven purely by currency weakness.

4 Strategic risk is highly concentrated. Palm oil, wheat, and sugar emerge as the most critical products, combining high cost, high dependency, and global price volatility.

5 Even a modest 25% reduction in high-dependency imports could save over 12,000+ crore BDT annually. However, the biggest cost drivers are not always the easiest to substitute in the short term.